

These programs are not your run-of-the-mill algorithms that execute situational choices based on a simple If/Then scenario. With the wealth of data that would become available, we need applications and software that use intelligent predictive algorithms which can foresee patterns and problems based on present and past behaviour. If such an application can use the sensor in the car to look at the tyre's wear and tear, factors in the weather forecast, takes into account real-time traffic updates and pulls the driver's alertness from his health band – it could make the decision to not start the car for safety reasons, since it's likely that the bad weather, traffic build-up, worn tyres and lack of sleep would result in an accident. The potential power of predictive intelligence from applications on the Internet of Things is unimaginable.

As the three steps of proliferation, connection and intelligent application is refined and implemented, we can expect the Internet of Things to be online in no time. But this is still a tentative framework which requires deep thought and structuring due to its immense investment in hardware, application development and most importantly, convincing the users that such deep penetration of technology in their lives will lead to a beneficial outcome.

Deconstructing an intelligent thing

Understanding a “thing” is easier than truly understanding what is an “intelligent thing”. In theory, we can say that any technological device that exhibits intelligent behaviour is an intelligent thing but in the world of the Internet of Things this is still too futuristic. An intelligent thing under these circumstances is usually a larger device that utilises an assortment of different sensors for data gathering and solution building. A simple example of such an intelligent thing is a farming harvester – this vehicle would have sensor devices in its wheels, brakes, radar system, self-driving system, weather sensors, moisture sensors, engine, storage tank and other places. The combination of these sensors would allow the harvester to factor in all the data from these sensors and process it using an advanced customised application. The purpose of the application would be oriented towards making the harvester a more efficient and helpful agricultural tool.

By taking into account all the data values, the sensors in the intelligent harvester would be able to regulate internal and external tasks. From knowing when the storage tank requires unloading to deciding which routes to take across a rain drenched field. The more the number of interconnected sensors in one thing, the greater its potential to be intelligent.